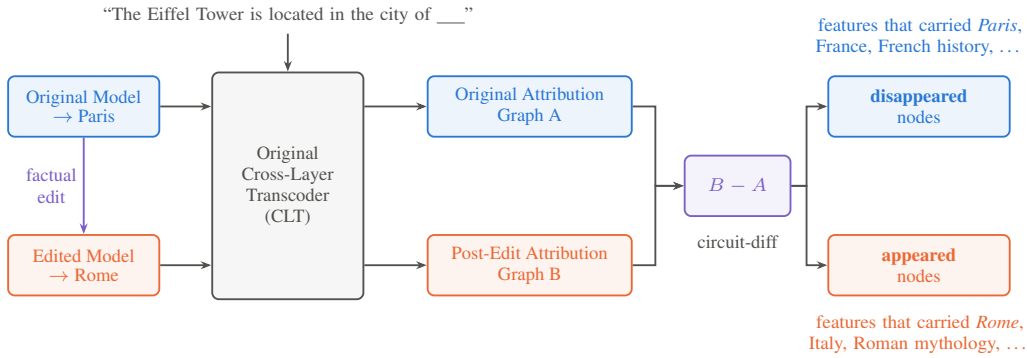


Circuit-Diff: Factual Edit-based Intervention Method for Localizing Knowledge in Attribution Graphs

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Abstract

Mechanistic interpretability defines *features* as the fundamental units of a neural network and *circuits* as the weighted subgraphs that carry out its computation. Because individual neurons are polysemantic, Cross-Layer Transcoders (CLTs) were introduced as a way to approximate a model’s circuits by generating an *attribution graph*. The nodes of that graph, however, are unlabeled features: reading a graph means pruning it and then working out by hand what each surviving node means. To make CLTs easier to use for circuit discovery, we introduce *Circuit-Diff*, which intervenes on the model itself with a low-rank factual edit and takes the features whose role in the attribution graph changes under that edit as related to the edited knowledge. On the edits we examine, the flagged nodes are not only detectors of the object token: read off the CLT’s released feature dashboards, they include features for the history, geography and associations surrounding the old and new objects. We formalize the method, measure how reliable a frozen CLT remains after a factual edit, test the selected nodes causally by patching them on up to 24 CounterFact edits, give a case study, and release an open-source implementation built on the *circuit-tracer* package Hanna et al. [2025], together with two further tools (multi-prompt aggregation and rule-based supernode labeling).

1 Introduction

Early mechanistic interpretability work defined *features* as the fundamental units of a neural network and *circuits* as the weighted subgraphs that represent the computation of the model Olah et al. [2020].

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Yet, studies found that neurons themselves did not correspond to single features, and instead existed in a state of superposition, in which polysemantic neurons would represent many unrelated features (Arora et al. [2018], Elhage et al. [2022], Olah et al. [2020]). As models grew larger and their internals harder to interpret, Sparse Autoencoders (SAEs) Cunningham et al. [2023], Bricken et al. [2023] and Cross-Layer Transcoders (CLTs) Persic et al. [2025] were introduced to approximate such features and circuits.

However, while SAEs and CLTs proved useful for approximating features and circuits, they are unsupervised and so do not by themselves ascribe semantic meaning to what they find. Early methods relied on manual inspection of where features fired [Bricken et al., 2023, Cunningham et al., 2023], and automated interpretability methods were introduced to generate descriptions of features [Bills et al., 2023, Paulo et al., 2024]. These descriptions have been found to be causally unfaithful under interventions in some settings [Huang et al., 2023]. In the CLT case, the generated *attribution graph* that approximates the circuit can be pruned, but a large number of nodes still need to be labeled. As CLTs and attribution graphs are adopted as a basis for circuit analysis across domains [Yang et al., 2026, Kim et al., 2026, Mazur et al., 2026], there is a need for tools that make this kind of analysis easier and more automated.

We introduce Circuit-Diff, a method that applies a causal intervention to a model via a low-rank knowledge edit using MEMIT [Meng et al., 2023a,b] and computes the difference in the attribution graphs Dunefsky et al. [2024], Persic et al. [2025] generated by a CLT trained on the unedited model and held fixed on both. The method surfaces feature nodes whose role in the attribution graph changes under the edit, and on the edits we examine those nodes read as the knowledge the edit moved. In this paper, we:

1. Formalize the *Circuit-Diff* algorithm, and describe two accompanying tools that we release but do not evaluate here: multi-prompt aggregation and, following Birardi and Paulo [2026], rule-based aggregation of nodes into supernodes, which we do not evaluate here.
2. Measure how reliable a frozen CLT remains after a factual edit, and how many feature nodes change role across $N = 500$ CounterFact edits.
3. Run ablation and steering tests (8–24 CounterFact edits) to check causal verifiability, finding that the selected nodes move the target logit substantially more than count-matched activation-ranked or random sets; on the inject side they also lead a top-influence ranking of the whole graph, while on the suppress side that ranking is as good or better.
4. Show a case study of the nodes the method identifies, read directly from the CLT’s released feature dashboards.
5. Release a library and local viewer for applying the method, built on circuit-tracer Hanna et al. [2025].

2 Related Work

Patel et al. [2026] take per-feature descriptions produced by existing auto-interpretability methods (Bills et al. [2023], Paulo et al. [2024]) and passes them directly to a language model, which groups them into supernodes. However, relying on auto-interpretability descriptions inherits the faithfulness issues Huang et al. [2023] report for such methods on SAEs. Arora et al. [2026] introduce ADAG, which characterizes each circuit component by its gradient-based attributions to input tokens and contributions to output logits, clusters functionally similar components into supernodes, and generates and scores natural-language labels for each group with a language model; the pipeline is demonstrated on attribution graphs over MLP neurons but is agnostic to the underlying unit basis. While ADAG replaces the descriptive signal used for grouping, its labels are still produced and scored by language models.

Birardi and Paulo [2026] avoid model-generated descriptions altogether: they group features by measuring each feature’s activation across a small set of concept-targeted probe prompts and clustering features with similar activation signatures under deterministic rules, validating the resulting supernodes with large-scale interventions. Chen et al. [2026] reduce the number of features prior to interpretation, scoring CLT features by gradient-based attribution to a task metric and retaining the minimal subset that preserves the model’s behavior, before running auto-interpretability on the retained set and evaluating both behavioral fidelity and explanation quality.

Our method shares the goal of Chen et al. [2026] (selecting which features warrant analysis before any interpretation occurs) and shares with Birardi and Paulo [2026] the commitment to grounding feature analysis in causal effects rather than generated descriptions. In fact, we adopt Birardi and Paulo’s procedure to group features by role into supernodes. We differ from both in the form of the intervention and in its position in the pipeline. Where Chen et al. [2026] construct their contrast at the input level, requiring a task-specific clean and corrupted dataset, we construct it at the weight level: a single low-rank factual edit [Meng et al., 2023a,b] specifies the counterfactual semantically, with the prompt held fixed. And where Birardi and Paulo [2026] use interventions to validate a grouping produced upstream, our intervention is the discovery signal itself: the features of interest are those whose role in the attribution graph changes under the edit.

3 Preliminaries

Factual edits. We edit with MEMIT Meng et al. [2023b], the multi-layer successor of ROME Meng et al. [2023a], on their *CounterFact* dataset. Each record is a fact (s, r, o) (subject, relation, object) with a prompt p that elicits o from s ; an edit request $(s, r, o \rightarrow o^*)$ asks the model to produce a counterfactual object o^* of the same relation on p instead. Each record also supplies *paraphrase* prompts (rephrasings of the same fact) and *neighborhood* prompts (different subjects sharing r and o), which we use to measure generalization and locality of the edit and, in §4.1, as the edit’s own null.

Attribution graphs. A cross-layer transcoder (CLT) T Persic et al. [2025] replaces the MLPs of a model M with transcoders Dunefsky et al. [2024] to yield a sparse dictionary of features whose decoders write to every later layer. Running M on a prompt p through T yields an attribution graph $G(M, T; p)$ Dunefsky et al. [2024], Persic et al. [2025]: a DAG whose nodes are token embeddings, active features, logits, and per-layer *error* nodes (the part of each MLP output T cannot reconstruct), with edges $w_{n \rightarrow n'}$ giving direct linear attributions between them. We write a feature node as $n = (i, t)$ for feature i at token position t , with activation a_n and *influence* I_n : the total normalized path weight from the target logit (o in the unedited graph, o^* in the edited one) back to n , summed over paths of every length on the row-normalized graph Persic et al. [2025]. Unlike the standard attribution graph method, which seeds this computation with the probabilities of all top logits, we seed a single target logit at unit weight, so I_n is the share of that logit’s attribution flowing through n and is comparable across the two graphs.

4 Circuit-Diff Method

4.1 Circuit-Diff Algorithm

Following the notation of §3, given a language model M_0 and a Cross-Layer Transcoder T_0 , we apply an edit request $(s, r, o \rightarrow o^*)$ to obtain a model M_e that elicits o^* instead of o on prompt p . We then compare the attribution graphs $G(M_0, T_0; p)$ and $G(M_e, T_0; p)$ to find feature nodes and edges associated with o and o^* . Throughout we use the running example of §6.1: moving the Eiffel Tower from Paris to Rome, i.e. (Eiffel Tower, located in city, Paris $\rightarrow$ Rome) on the prompt “is located in the city of”.

We split our process into five steps:

Step 1: Perform the fact edit. We use MEMIT [Meng et al., 2023b] to apply the edit request to M_0 , producing a model M_e that, when the edit succeeds, elicits o^* (here Rome) on p . We prefer multi-layer MEMIT to single-layer ROME [Meng et al., 2023a] because factual associations are reported to be distributed over several layers [Geva et al., 2023, Haviv et al., 2023, Nichani et al., 2024, Persic et al., 2025]; see Appendix F.

Step 2: Compute the two attribution graphs and their difference. Given M_0 and M_e , we compute $A = G(M_0, T_0; p)$ and $B = G(M_e, T_0; p)$, using the same CLT on both so that node identities are comparable, an assumption §5.1 tests and §7 qualifies. For every node $n = (i, t)$, i.e. T_0 ’s feature i at token position t , we compute the change in the node’s influence on the target logit (o ’s logit node in A , o^* ’s in B) and δ_n , which tracks the normalized movement of its activation. For each edge we compute the change in edge weight and an influence-weighted distribution over n ’s outgoing edges. All quantities are defined in Table 1.

Table 1: Per-node difference quantities extracted from the attribution-graph pair $A = G(M_0, T_0; p)$ and $B = G(M_e, T_0; p)$, for each node $n = (i, t)$ (feature i at token position t) and edge $n \rightarrow n'$.

Quantity	Definition	Measures
Δa_n	$a_n^{(B)} - a_n^{(A)}$	Change in activation of feature i at token position t .
ΔI_n	$I_n^{(B)} - I_n^{(A)}$	Change in node n 's influence on the target logit (o^* 's logit node in B , o 's in A).
$\Delta w_{n \rightarrow n'}$	$w_{n \rightarrow n'}^{(B)} - w_{n \rightarrow n'}^{(A)}$	Change in edge weight from node n to node n' .
$\pi_n(n')$	$\propto w_{n \rightarrow n'} \cdot \max(I_{n'}^{(A)} , I_{n'}^{(B)})$	Importance-weighted distribution of n 's outgoing edge mass, normalized over its targets;
δ_n	$\Delta a_n / \max(a_n^{(B)}, a_n^{(A)}, \varepsilon_n) $	Normalized activation movement, in $[-1, 1]$; the guard $\varepsilon_n = \tau_a \cdot a_i^{\max}$ (feature-level maximum of n 's feature i) keeps near-silent nodes from saturating the ratio.

Step 3: Track appearing and disappearing nodes. We use δ_n from Table 1 to characterize each feature's behavior under the diff. A node n **increased** if $\delta_n > 0$ and **decreased** if $\delta_n < 0$. The extremes are the candidates of interest: a node **disappeared** if $\delta_n = -1$, making it a candidate for a node associated with predicting o , and **appeared** if $\delta_n = 1$, making it a candidate for a node associated with predicting o^* .

Step 4: Apply a null calibration. As §5.2 shows, a large fraction of nodes fall into the appeared and disappeared sets, so a further filter is useful when a more constrained set is wanted. We compute a floor $\epsilon = Q_p(\{|a_n^{(B)} - a_n^{(A)}| : n \in \text{NBR}\})$ from the neighborhood prompts (which elicit unedited knowledge and so serve as the edit's own null), where p is the chosen percentile, and keep on each side only the nodes whose movement exceeds ϵ . Every ablation in §5.3 is reported both with and without this floor.

Step 5: Rank the nodes on each side. Since we hypothesize that increased and decreased nodes correspond to o^* and o respectively, we split the nodes into two rankings, one per behavior, and score each with a heuristic meant to surface the features most associated with the change. We rank each side with $\phi_n^{(a)} = |\delta_n|^\alpha |\Delta I_n|$, the weighted logit change (α per Appendix A), and measure it against two method-free reference baselines over the same nodes: activation, $\phi_{\text{act},n} = |a_n^{\text{ref}}|$ (the activation λ scales), and influence, $\phi_{\text{inf},n} = \max(|I_n^{(A)}|, |I_n^{(B)}|)$. All are unsigned ranking strengths; direction comes from $\text{sign}(\delta_n)$, which splits the nodes into appeared and disappeared before ranking. Five further scores we tried are defined in Table 3; §5.3 finds four of them indistinguishable from $\phi^{(a)}$ and one, $\phi^{(e)}$, no better than random.

4.2 Circuit-Diff Platform and Tools

Along with this paper, we release the Circuit-Diff Platform, which builds on the circuit-tracer package Hanna et al. [2025] to compute the diff, inspect it visually and export the tables and figures used here (Appendix K). Two further tools ship with it and are not evaluated here: *multi-prompt aggregation*, which pools a feature's score over rephrasings of the same fact (Appendix K.1), and *supernode labeling*, which sorts the returned features into roles by their activations on a fixed probe set, with no language-model-written descriptions, following Birardi and Paulo [2026] (Appendix G).

5 Experiments

All experiments use Gemma-2-2B [Team et al., 2024] and the released 426k-feature CLT of Piotrowski and Hanna [2025]. We refer to four model-CLT combinations on a given fact: A and B from §4.1, plus the adapted-lens graphs $C = G(M_e, T_f; p)$ and $D = G(M_0, T_f; p)$, where T_f is the CLT fine-tuned on M_e (Appendix H). Each pair isolates one factor: $B - A$ is the circuit-diff; C versus B isolates lens adaptation with the model fixed at M_e ; and D is the control corner: the fine-tuned

lens on the *unedited* model, i.e. what fine-tuning alone does absent any edit. Compute is reported in Appendix B.

5.1 Cross-Layer Transcoders Reliability After Factual Edit

To test whether T_0 survives the edit we compare the completeness score, the replacement score and per-layer cosine activation similarity between A and B over the edit, paraphrase and neighborhood prompts of §3 (Figure 1). The damage to the lens is localized to the edit. The replacement score of the attribution graph falls sharply on edited prompts (median drop ~ 0.3) while the less strict completeness score falls much less (median drop ~ 0.06); on neighborhood prompts neither moves appreciably. Because the drop appears on the edit and paraphrase buckets but not on the neighborhood bucket, we attribute it to the knowledge edit, and read the gap between the two scores as the CLT’s error nodes absorbing what the lens can no longer model. This remains a core limitation (§7). We deliberately do not fine-tune the base CLT, since doing so would break the correspondence between its features and the released feature dashboards we read them from.

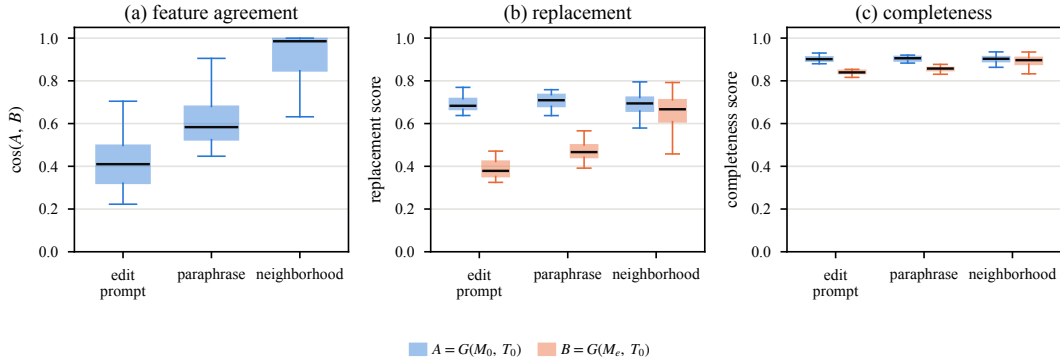


Figure 1: A single MEMIT edit damages the frozen lens exactly where it lands; while the replacement score drops, the less strict completeness score isn’t as low in the edited case. Per-prompt agreement and graph quality for $A = G(M_0, T_0)$ versus $B = G(M_e, T_0)$. A single fact is edited per model ($k = 1$) over $n = 20$ trials; within each trial we split by prompt bucket, where edit and paraphrase both elicit the edited fact and neighborhood elicits unedited facts.

We also fine-tuned the CLT on M_e ’s activations over general wiki text (T_f); the fine-tune early-stops at parity and does not recover the replacement score, so the frozen lens T_0 is used throughout (Appendix H).

5.2 Circuit-Diff Nodes that Appeared/Disappeared

How much of a graph does one edit move? Figure 2 answers this over $N = 500$ CounterFact edits under three readings of “changed”: direction, presence, and *passed null* (Step 4, §4.1). The first two are close to uninformative on their own: in panel (b) the gained side covers 96.2% of the graph by direction and 94.4% by presence, and the lost side 48.2% and 27.2%. Applying the floor collapses these to 14.0% and 3.2% respectively. Panel (c) shows the same effect against the influence ranking: unfloored, the increased share rises with q but stays below its chance level throughout, whereas floored it sits 2–3 $\times$ above its 14.0% chance rule at every $q < 100\%$.

5.3 Verification of Circuit-Diff Nodes

Patching protocol. To verify the selected nodes causally we patch them in the unedited model. Since each node also feeds later layers, we use Persic et al.’s constrained patching: given a node $n = (i, t)$, we clamp its value to $\lambda \cdot a_n$ and add it to the original MLP output for a window of layers ahead of it, sweeping the window endpoint. Each patch is applied to $G(M_0, T_0; p)$: appeared nodes are scored on whether they reproduce the effect of the edit (raising the o^* logit), disappeared nodes on whether they suppress the old answer (lowering the o logit). We score a patch by R^z , its effect as a fraction of the edit’s own on the centered-logit basis, so $R^z = 1$ reproduces the edit exactly ($\varepsilon_z = 0.5$ guards the denominator):

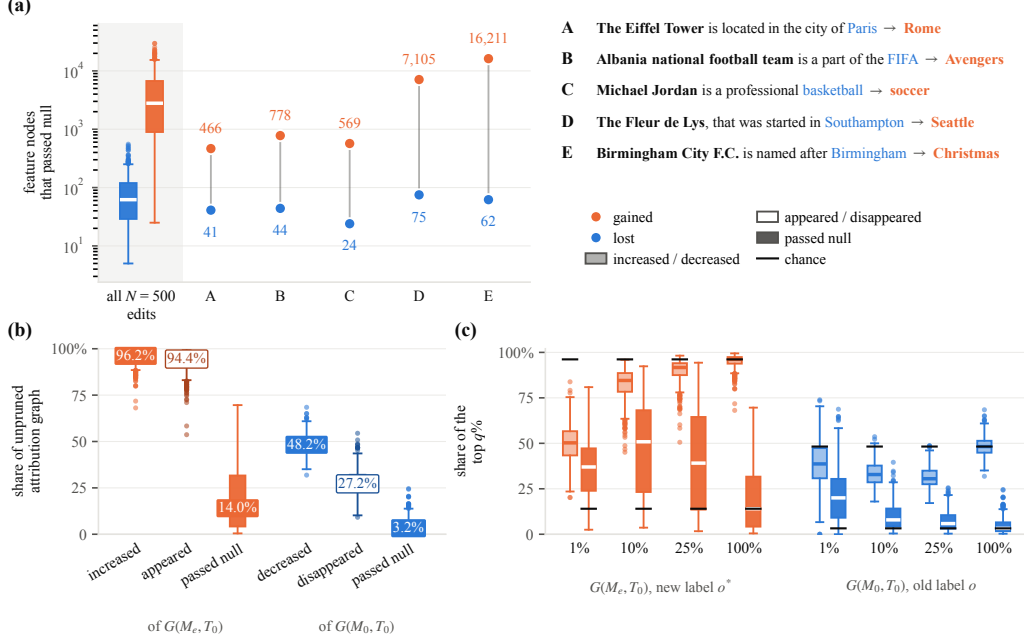


Figure 2: **Feature nodes changed by a knowledge edit, under three readings.** $N = 500$ Counter-Fact edits, $G(M_0, T_0)$ vs. $G(M_e, T_0)$ on the edited prompt. Direction (increased/decreased), presence (appeared/disappeared), and *passed null*: moved by more than the edit’s own control-prompt floor, the 99th percentile of raw $|\Delta a_n|$ (§4.1, Step 4). **(a)** Passed-null nodes per edit, each against its own floor, so counts are not comparable across edits; A is a case study, not among the N . **(b)** Each side as a share of its own graph. **(c)** Share of the top- $q\%$ by influence on the target logit, direction vs. passed null. The two sides use different rankings (gained: $G(M_e, T_0)$ against o^* ; lost: $G(M_0, T_0)$ against o) and are not comparable; black rules are each series’ chance level.

$$R_t^z = \frac{z_t^{\text{patch}} - z_t^{(M_0)}}{\max(|z_t^{(M_e)} - z_t^{(M_0)}|, \varepsilon_z)}, \quad z_t = \ell_t - \frac{1}{|V|} \sum_{v \in V} \ell_v. \quad (1)$$

Selection and floor policy. All arms rank with $\phi^{(a)}$ (§4.1, Step 5); at the depths we patch (top 10% at most) every selected node is an appeared/disappeared node, not merely an increased/decreased one. We report every ablation under both calibrations: unfloored, and restricted to the nodes that pass the $p = 99$ null floor of Step 4 (median survival 13% of the increased side and 8% of the decreased side); under the floor, q is a share of the survivors. Every arm is compared against two method-free baselines that rank *every* node of the graph by influence or activation on that side’s own target ($\phi_{\text{inf}}^{\text{graph}}$, $\phi_{\text{act}}^{\text{graph}}$) and against a random draw, with counts matched at every rung, so a gap between arms is which nodes were selected and never how many.

Single nodes. As Persic et al. [2025] report, one feature node rarely moves the logit on its own, because a concept is split across several features; the per-node sweep is in Appendix J (Figure 9). Per node, the whole-graph influence ranking is ahead of $\phi^{(a)}$ on the suppress side at every rung under both calibrations and on the unfloored inject side; only on the floored inject side at $q \geq 5\%$ does $\phi^{(a)}$ lead. The activation baseline falls to random once it cannot inherit the diff’s partition. These are pooled medians over 8 edits, and no per-edit sign test is available at the single-node level, so they should be read as descriptive.

Groups. To see the effect of the set the method returns, we patch the whole top- $q\%$ set in one pass (Figure 3). At $q = 5\%$ the selected set moves the target logit further than a count-matched whole-graph influence ranking on the inject side (9.98 versus 5.90 centered logits unfloored, 8.16

versus 4.94 floored); on the suppress side the two are comparable unfloored (-42.4 versus -41.3) and the baseline is ahead under the floor (-24.8 versus -35.3). Grouping by *supernode*, i.e. by shared meaning or role, would likely be a sharper unit of intervention than the top $q\%$ of a side taken as a whole; we leave that comparison to future work.

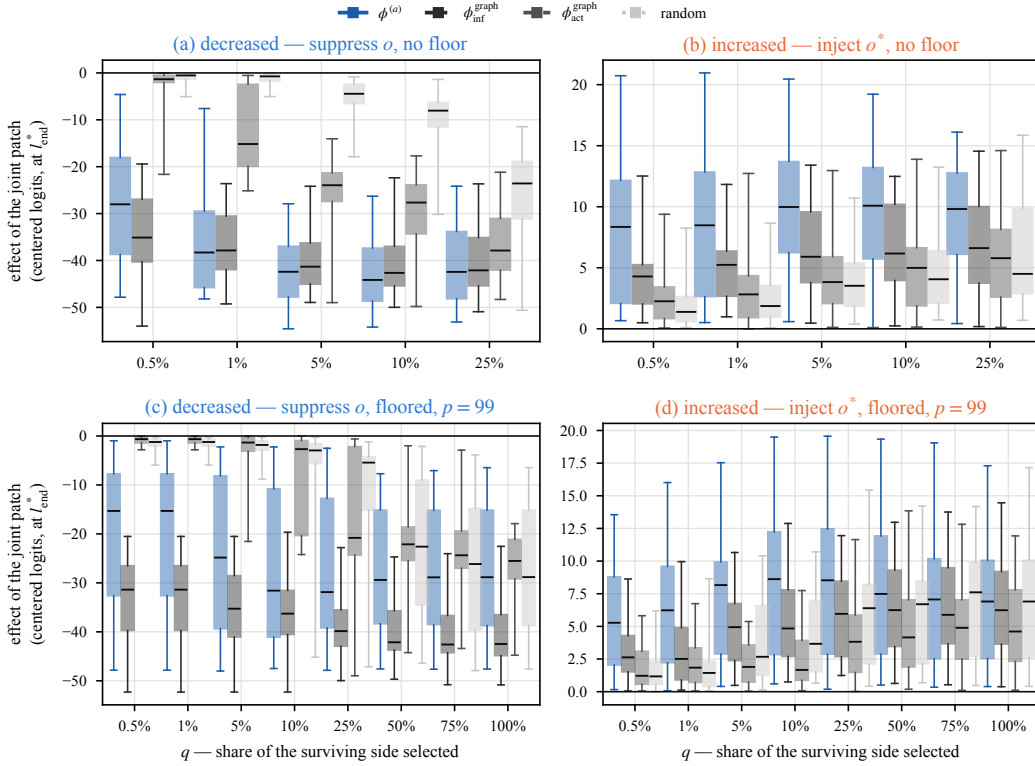


Figure 3: **Group ablation: the whole selected set patched in one pass, under both calibrations.** Every member of the top- $q\%$ set is clamped simultaneously; arms, count matching and the l_{end}^* criterion as in Fig. 9, each set read at its own best ($\lambda, l_{\text{end}}^*$). Top row: no floor. Bottom row: $p = 99$ survivors; its ladder runs to $q = 100\%$, where every arm patches the same survivor set and the arms coincide. Boxes are 24 CounterFact edits, box = IQR, whiskers = full range.

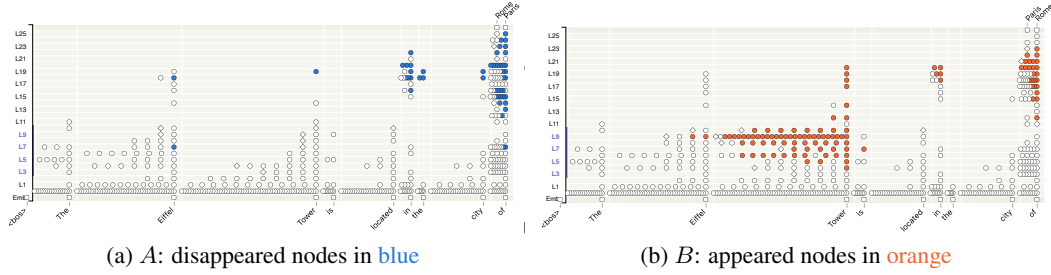
Which score. Widening the group ablation to all six scores (Table 3 and Figure 11, Appendix J) changes nothing: $\phi^{(a)} - \phi^{(d)}$ are indistinguishable because each multiplies influence by $|\delta_n|$, which is saturated at 1 for nearly every node on both sides, and $\phi^{(e)}$, the only edge-only score, tracks random. Against the whole-graph influence baseline, per-edit sign tests at the 0.156 replicate-noise threshold favor $\phi^{(a)}$ on the unfloored inject side on 9–11 of the 12 edits at every q ; on the suppress side the result is mixed, and at $q \leq 1\%$ the baseline is ahead.

6 Case Studies

Figure 4 lets a reader answer the question the method exists for, unaided: do the nodes circuit-diff flags *look like* the old and new objects, with no label written by us or by a language model? Every feature appears exactly as the CLT’s released dashboard Piotrowski and Hanna [2025] presents it (the tokens it promotes and its highest-activating context), with the top-15 per side in Appendix K.2.

6.1 Paris $\rightarrow$ Rome

The edit landed (M_e ’s argmax on p is Rome), though the frozen lens keeps a replacement score of only 0.49 on M_e , at the low end of the post-edit range in Figure 1. The two graphs differ where the edit predicts they should. Disappeared nodes concentrate at the final position in layers 15–25, the



(a) A: disappeared nodes in blue

(b) B: appeared nodes in orange

Top readouts onto “Paris” in A (base model)

#1 L18/f7269 +1.61 ϕ 1	... world (British Museum at London, Grand Palais at Paris, ...	le, É, Ré, dé, les
#2 L20/f9908 +1.23 ϕ 2	... sculpture can be seen at The Louvre Museum in Paris, ...	normaux, TagMode, financières, parsedMessage, sauvages
#3 L19/f9768 -1.06	University of Tokyo 7-3-1 Hongo, Bunkyo-ku, Tokyo 1...	úgó, bienven,
#4 L25/f9331 -1.03 ϕ 4	..., Ecole Normale Supérieure, Paris, France, 3...	tvguidetime, stratégique, nôtre, daction, Greif
#5 L1/f3872 +1.02	... Mifflin Harcourt Publishing Company. All rights reserved...	Etr, Aene, Caesar, Eocene
#6 L18/f14136 -0.92 ϕ 3	... world (British Museum at London, Grand Palais at Paris, ...	WithFormat, addCriterion, uska, ainfi, excellents
#7 L0/f7081 +0.84	<bos>Ce va s...	EndGlobalSection, IUrlHelper
#8 L19/f472 +0.79 ϕ 5 @in	... Liberated and enfranchised by the events of 1789, ...	departments, department, Giver, Als, Seine

Top readouts onto “Rome” in B (edited model)

#1 L19/f9768 -1.39	University of Tokyo 7-3-1 Hongo, Bunkyo-ku, Tokyo 1...	úgó, bienven,
#2 L17/f11572 +1.30	... participating from London to Sao Paolo, Seoul to Babylon...	Lisbon, Istanbul, Bangkok, London, Sydney
#3 L20/f4888 +1.23 ϕ 1	... at the Regina Elena National Cancer Institute in Rome, ...	HomeAsUp, Lati, +##, MENAFN, snippetHide
#4 L24/f8520 +1.16	... ratio of the moments of proton and deuteron by Bloch, ...	Jefus, Ans, several, myself, itself
#5 L0/f9908 +1.12	... makes a person feel sad and hopeless much of the...	myself, Jefus, small, purpose, several
#6 L1/f3872 +1.10	... Mifflin Harcourt Publishing Company. All rights reserved...	Etr, Aene, Caesar, Eocene
#7 L19/f4224 +0.96 ϕ 3	... Trajan and Marcia Furnilla (second wife of the...	Caesar, Fla, legion, Marcus, Roman
#8 L0/f8961 +0.93	... for HIV/AIDS patients. January Institute of Medicine...	Efq, Majefty, Jefus, Monfieur, purpose

Figure 4: What the readout ranking sees, and what the diff flags. (a, b) the two graphs of the running example. (c, d) the features with the largest direct weight $|w|$ onto the answer logit in each graph (sign shown; negative = suppresses), the ranking a circuit-tracer session gives with no diff. Each carries the dashboard’s highest-activating context (shaded by activation, peak in bold) and its top promoted logits. Features sit at the final token unless marked @token. Bar and shading carry the diff’s verdict: blue = disappeared, orange = appeared, gray = in both; ϕn is the rank under $\phi^{(a)}$ on that side.

readout of the old answer, and at the relation tokens “in the”; appeared nodes form a band over layers 7–10 at the subject’s last token, exactly where MEMIT writes, plus a new late-layer column at the final position.

The top movers are recognizable without a label (Figure 4c, d; Appendix K.2). The rank-1 disappeared node L18/f7269 promotes French tokens and fires on “Grand Palais at Paris”; L20/f9908 fires on “Louvre Museum in Paris”. The rank-1 appeared node L20/f4888 fires on “in Rome, Italy”-style contexts, and L17/f13216 (rank 4, sitting at the subject token) promotes “Italy”: the subject representation itself now carries the new object. The same reading holds on the other edits we inspected: two of them in §6.2 below, and, for one more, the French → English edit loses “françaises / franceses” and “Laurent / Sarkozy” features.

Beyond the token. On this edit the flagged features are not only detectors of the answer string; several of them read as the surrounding idea of the place. Deeper in the appeared ranking sit a Roman-antiquity feature promoting “Caesar / legion / Roman” (L19/f4224, rank 3), a Roman-mythology feature firing on “Romulus and Remus in 753” (L20/f7760, rank 5) and a papacy feature firing on “rival popes in Rome and Avignon” (L22/f4296, rank 13); the disappeared side mirrors it with a French Revolution feature (“the events of 1789”; L19/f472, rank 5) and French port cities (“Havre /

Lyons / Nice”; L19/f12469, rank 14). Four more of each are in Appendix K.2. What the diff surfaces on this edit is therefore Roman history, mythology and the Church on one side and French history and geography on the other, rather than the two object tokens alone.

Recurrence separates knowledge from mechanism. Some top features recur across edits; the dagger counts in how many of the 25 (the running example plus the 24 of the sweep) a feature appears. L18/f7269^{†4} recurs in exactly the other French facts of the sweep, recurrence that tracks shared knowledge. By contrast L7/f12622^{†12}, L5/f8658^{†5} and L25/f9787^{†11} recur across a dozen unrelated edits with uninterpretable dashboards, and the first two sit at the subject position inside the MEMIT band, which suggests they track the write itself rather than the fact. A reader should discount them, and the platform reports a cross-edit recurrence count as a cheap filter for exactly that (Appendix K.2).

6.2 Two further edits

We apply the same reading to two further edits from the sweep of §5.3: *Taj Mahal, follower of, Islam* → *Christianity* and *Pentium II, developed by, Intel* → *Microsoft*. Both repeat the geometry of §6.1 (appeared nodes banded across the MEMIT layers at the subject’s last token, disappeared nodes columned at the final position), and both name their objects without a label: Taj Mahal trades a “Shah / Allah” feature for “Protestant / Christian / churches”, and Pentium II trades a “Hyper Threading enabled CPU” feature for one firing inside Transact-SQL documentation. Both diffs and their top-6 features per side are in Appendix K.3.

7 Limitations

Dependence on the edit and on the lens. The method rests on two things: that the factual edit succeeded, and that the base CLT remains a usable lens afterward. The weaker the edit, the less we expect the diff to mean: if the model barely changed, few features move, and the ones that do need not be the ones carrying the fact. We do not measure this dependence directly.

The lens is the more fundamental limitation. Holding T_0 fixed preserves the meaning its features had for the base model, which is what makes the released dashboards readable, but does not guarantee that meaning still describes M_e . §5.1 shows high pre/post agreement away from the fact, and the replacement–completeness gap is consistent with error nodes absorbing what the lens can no longer model; but that is an inference from two aggregate scores, and it bounds how much of the knowledge-carrying circuit the method can recover.

Practicality and scale of the evidence. Figure 2 shows that the raw appeared/disappeared reading is too permissive to be useful on its own: it covers most of the graph on the gained side. The null floor of Step 4 is what makes the output a short list (14.0% and 3.2% of each side), so the method’s practical value rests on a calibration whose percentile we have not tuned systematically. Single-node patches move the logit little, as Persic et al. [2025] also report, so the causal evidence rests on group patches over 24 edits (12 for the six-score comparison), which is enough for a per-edit sign test but not for a tight effect estimate. And the selected sets overlap heavily with the graph’s top-influence set, so on the suppress side the advantage over simply reading off influence is small or absent.

Limitations inherited from the edit method. Causal tracing and ROME/MEMIT carry known problems that the diff inherits. Causal tracing does not reliably identify the best MLP layer at which to overwrite a fact [Hase et al., 2023]; longer-form evaluation of edited models finds the edit applied inconsistently [Rosati et al., 2024]; and knowledge editing produces unwanted side effects at scale [Hoelscher-Obermaier et al., 2023, Yang et al., 2024]. Any such side effect appears in the diff as movement that is not the fact, which is one reason the recurrence filter of §6 is worth reporting alongside the ranking.

Limitations of the CLT basis itself. Cross-Layer Transcoders are now a common tool for approximating circuits in language models, but their mechanistic faithfulness has been questioned on toy models where ground truth is available [Lange et al., 2026]. Circuit-Diff inherits whatever the basis gets wrong: it can only report movement among features the dictionary happens to represent.

8 Conclusion

We set out to make the feature nodes that carry a fact easier to find, and the simplest version of the idea already carries signal: edit the fact, rebuild the attribution graph through the same frozen CLT, and look at what moved. On the 24 edits we patch, we find that the features our method elicits present, checked from their own dashboards and ablations, as the knowledge around the object rather than just its name. We release the platform and every table behind these figures so others can try the method on their own edits and CLTs.

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A Defaults Used in Experiments

Unless a deviation is stated where an experiment is described, all results use the defaults in Table 2.

Table 2: Defaults used across experiments.

Parameter	Default	Role
<i>Scores (§4.1 and Table 3)</i>		
α	1.0	exponent in $\phi^{(a)}, \phi^{(b)}$ (sweep: $\{0.5, 1, 2\}$)
τ_a	0	activation guard off; $\frac{0}{0} \equiv 0$ in δ_n
MAD floor	10^{-12}	denominator floor in $\phi^{(c)}$'s robust z
<i>Feature-patching verification</i>		
ε_R	0.01	guard for $R_{\text{old}}, R_{\text{new}}$
λ grid	$\{0.5, 1, 2, 4, 6, 8\}, -\{0, 1, 2, 4, 6, 8\}$	appeared, disappeared
λ^*	$\arg \max R_{\text{new}}^{\text{adj}} / \arg \min R_{\text{old}}^{\text{adj}}$	pre-registered per node
q grid	$\{0.1, 0.5, 1, 5, 10\}\%$	selection quantiles
KL ceiling	1.0 nat	rows above excluded
<i>Editing and CLT</i>		
$\mathcal{B}$	$[3, 10]$	MEMIT edit band
k	1	facts per model
η	10^{-4}	CLT fine-tune LR

B Compute Resources

Every experiment in this paper runs on a single GPU. Attribution-graph computation and patching use one NVIDIA RTX A6000 (48 GB) on an internal SLURM cluster, with ~ 96 GB of host memory used to stage the 426k-feature CLT before it is moved to the device; the 48 GB budget is the binding constraint, and is what forces the decoder-slice paging described in Appendix H. MEMIT edits and prompt resolution are comparatively cheap and run in the same allocation. No experiment here requires multi-GPU or multi-node execution, and no model is trained from scratch: the only training is the additive-delta CLT fine-tune of Appendix H, which at $k = 1$ stops after 100 optimizer steps (181,454 tokens). The dominant cost is the patching sweep. The 24-edit group ablation of §5.3 measured 154,345 patch cells for a total of 10.3 GPU-hours, a median of 0.42 GPU-hours per edit ($\sim 6,300$ cells each), each edit fitting in a single shard. The full project consumed substantially more compute than the runs reported here, through preliminary sweeps, discarded floor percentiles and score variants, and the multi-edit ladder of Appendix I.

C Released Code and Data

The Circuit-Diff library is released as a pip-installable package (`pip install circuit-diff`, `import circuit_diff`, CLI `cs`) at <https://github.com/edwardF8/Circuit-Diff>, under the MIT license. It contains the diff computation, the node scores of §4.1, the patching stages used in §5.3, the multi-prompt aggregation of Appendix K.1, and a local viewer. The viewer vendors a patched copy of Anthropic’s attribution-graphs frontend (MIT, © 2025 Anthropic), with the patches documented in the repository. A bundled 11 MB attribution bundle for the Eiffel Paris $\rightarrow$ Rome case study ships with the package, so `cs load <bundle> -serve` reproduces Figure 4(a, b) with no computation. The public adapter targets Gemma-2-2B only. A hosted demo of the viewer runs at <https://circuit-diff-paper.vercel.app/>. A second branch, `experiments`, adds the campaign code, run configurations and figure scripts behind the quantitative figures (`setdiff-500`, `cf-sweep-24`, `ablation-redo`, `ablation-verification`, `score-ablation`).

The summary tables behind those figures are released separately as a dataset (23 MB, 83 files) at <https://huggingface.co/datasets/edwardF8/Circuit-Diff-paper-data> under CC-BY-4.0: the full Eiffel attribution bundle, the per-edit set-difference and null-floor tables behind Figure 2, the 24-edit CounterFact sweep, the group and single-node patching tables behind Figures 3, 9 and 10 including the null-floor and duplicate-gate robustness variants, the rank-causality dose-response tables, and the six-score comparison behind Figure 11.

What is not released. No model weights and no edited checkpoints. The MEMIT weight deltas for the case-study edits are not part of this drop. Base weights come from Google under the Gemma Terms of Use.

Reproducibility boundary. Loading, viewing and ranking a released bundle, and regenerating the paper’s diff, patching and case-study tables from the released data, require none of the editing stack: the library’s load path depends only on `pandas` and `pyarrow`. Computing *new* bundles or edits does require a GPU, gated access to `google/gemma-2-2b`, `circuit-tracer` 0.4.1 (decoderresearch fork) and MEMIT wiring that is not part of this release. The MEMIT-reproduction figures (Appendix F) and the CLT lens-grid figures (Appendices H and I) come from a separate experimental codebase and are not regenerated by the released repository.

D Assets and Licenses

All artifacts used in this paper are publicly released and are used under their stated terms. Gemma-2-2B [Team et al., 2024] is used under the Gemma Terms of Use. The CounterFact dataset and the ROME/MEMIT implementation [Meng et al., 2023a,b] are used under the MIT license. `circuit-tracer` [Hanna et al., 2025], on which our platform is built, is used at version 0.4.1 (decoderresearch fork) under the MIT license. The 426k-feature cross-layer transcoder for Gemma-2-2B and its feature dashboards [Piotrowski and Hanna, 2025] are used under the terms stated on the model’s Hugging Face card. The probe-prompting role assignment of Appendix G is adapted from Birardi and Paulo [2026], with the deviations from their published procedure noted there. The Circuit-Diff platform is released under the MIT license and its accompanying tables under CC-BY-4.0; the release vendors a patched copy of Anthropic’s attribution-graphs frontend (MIT, © 2025 Anthropic). No model weights, edited checkpoints or new datasets are released with it (Appendix C).

E Broader Impacts

Circuit-Diff is a tool for locating the features a fact runs through. The intended use is auditing and debugging: knowing where a model stores a claim is a prerequisite for checking whether it stores it at all, for tracing an incorrect output to its source, and for evaluating whether an edit did what its author intended. The same localization could in principle make targeted modification of a model’s factual behavior easier, including modification a downstream user would not notice; we note that the localization signal here is obtained *from* an edit, so it does not lower the barrier to performing one.

The risk specific to this class of method is subtler than misuse: a localization can be wrong and still look convincing. A ranked list of features with readable dashboards invites the reader to believe a circuit has been identified when what has been identified is movement in an approximation. We have tried to make that failure mode visible rather than hide it, by reporting how much the frozen lens degrades after an edit (§5.1), by showing that the unfloored reading covers most of the graph (§5.2), and by reporting the side on which a plain influence ranking matches or beats our score (§5.3). Interpretability results of this kind should not be used to make claims about what a deployed model does or does not know without independent verification.

F MEMIT Reproduction on Gemma-2-2B

We perform factual edits with ROME [Meng et al., 2023a] and its successor MEMIT [Meng et al., 2023b]. Meng et al. [2023a] introduce causal tracing, a series of causal interventions on a model’s internal activations that identifies which layers and components are decisive for a factual prediction. Figure 5 shows our reproduction of those causal-tracing results on Gemma-2-2B; following their convention we edit the layer band $\mathcal{B} = [3, 10]$, which has the largest gap between the effect of severing the MLP and severing the attention modules. ROME and MEMIT apply low-rank updates to the feed-forward MLPs of the layers in $\mathcal{B}$ to change the stored association. MEMIT scales to large batches of edits, but §5.1 and Appendix I show that the frozen transcoder degrades sharply as the number of edits grows, so we edit one fact per model throughout.

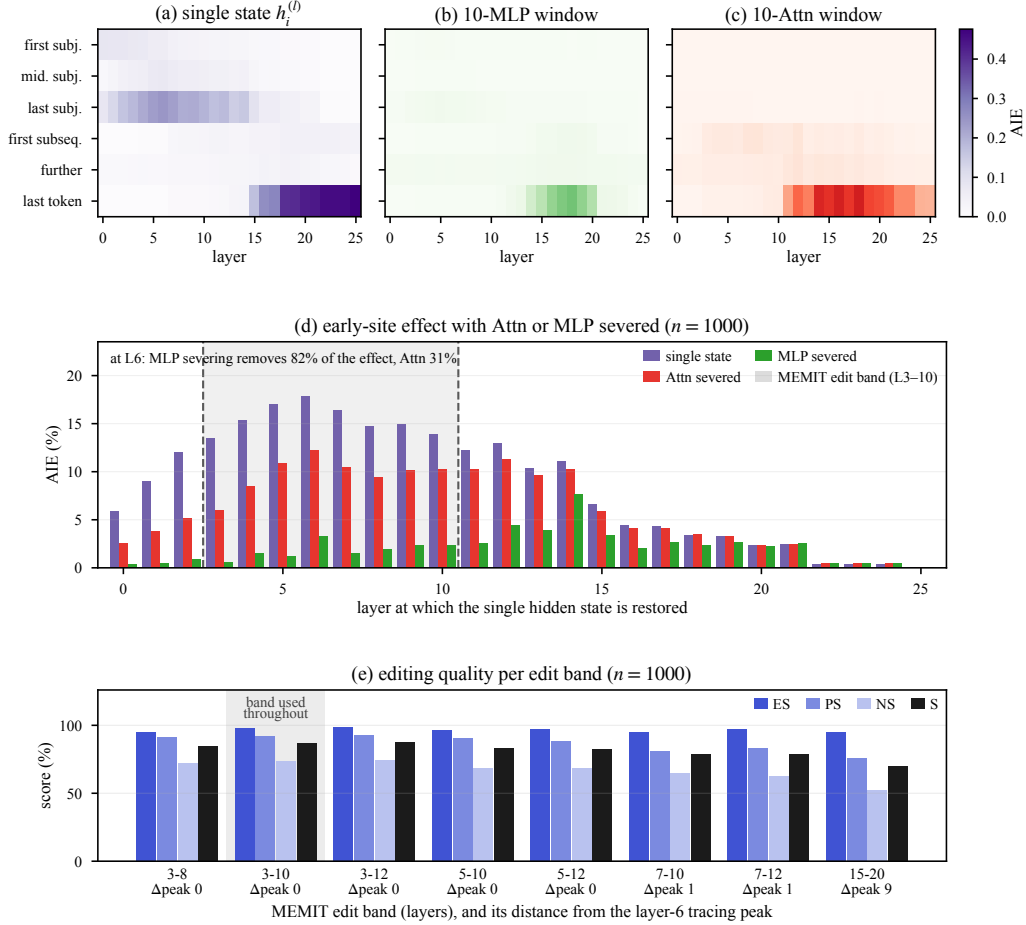


Figure 5: Causal tracing on Gemma-2-2B, and the layer-band ablation used to select the edit layers. Panels (a)–(c) reproduce the averaged causal-trace heatmaps of Meng et al. [2023a], panel (d) reproduces the severed-module variant from the same work, and panel (e) reproduces Figure 11 of Meng et al. [2023b]. (a) Average indirect effect (AIE) of restoring a single clean hidden state in a run where the subject tokens have been corrupted with noise. (b) The same measurement for restoring a window of ten consecutive MLP outputs, with the window center on the horizontal axis. (c) The same for a window of ten attention outputs. Panels (a)–(c) share a single color scale, set by panel (a), following the upstream figure. (d) AIE at the last subject token when either all MLP or all attention modules at that token are held at their corrupted values. Shading marks layers 3–10, the layer choice used for all MEMIT edits in this paper. (e) Editing metrics for eight candidate bands, with MEMIT applied to the same 1000 CounterFact records in each case, see Figure 6 for the definition of each metric.

G Probe-Prompting Details

All signature statistics and thresholds follow Table 3 of Birardi and Paulo [2026]. For feature i and probe x , $\hat{a}_i(x)$ is the peak activation over non-BOS positions and a probe is active when $\hat{a}_i(x) > 0$. Per feature we compute the peak consistency γ_i (fraction of probes containing the modal peak token in which it is the peak), the number of distinct peak tokens, the functional-token share c_{func} over active probes ($c_{\text{sem}} = 1 - c_{\text{func}}$), the functional-versus-semantic margin $\Delta_{\text{fs},i} = 100 (\hat{a}_i^{\text{func}} - \hat{a}_i^{\text{sem}}) / \max(\hat{a}_i^{\text{func}}, \hat{a}_i^{\text{sem}})$, and the median sparsity $\tilde{\text{sp}}_i$ of $\text{sp}_i(x) = (\hat{a}_i(x) - \bar{a}_i(x)) / \hat{a}_i(x)$. Roles are

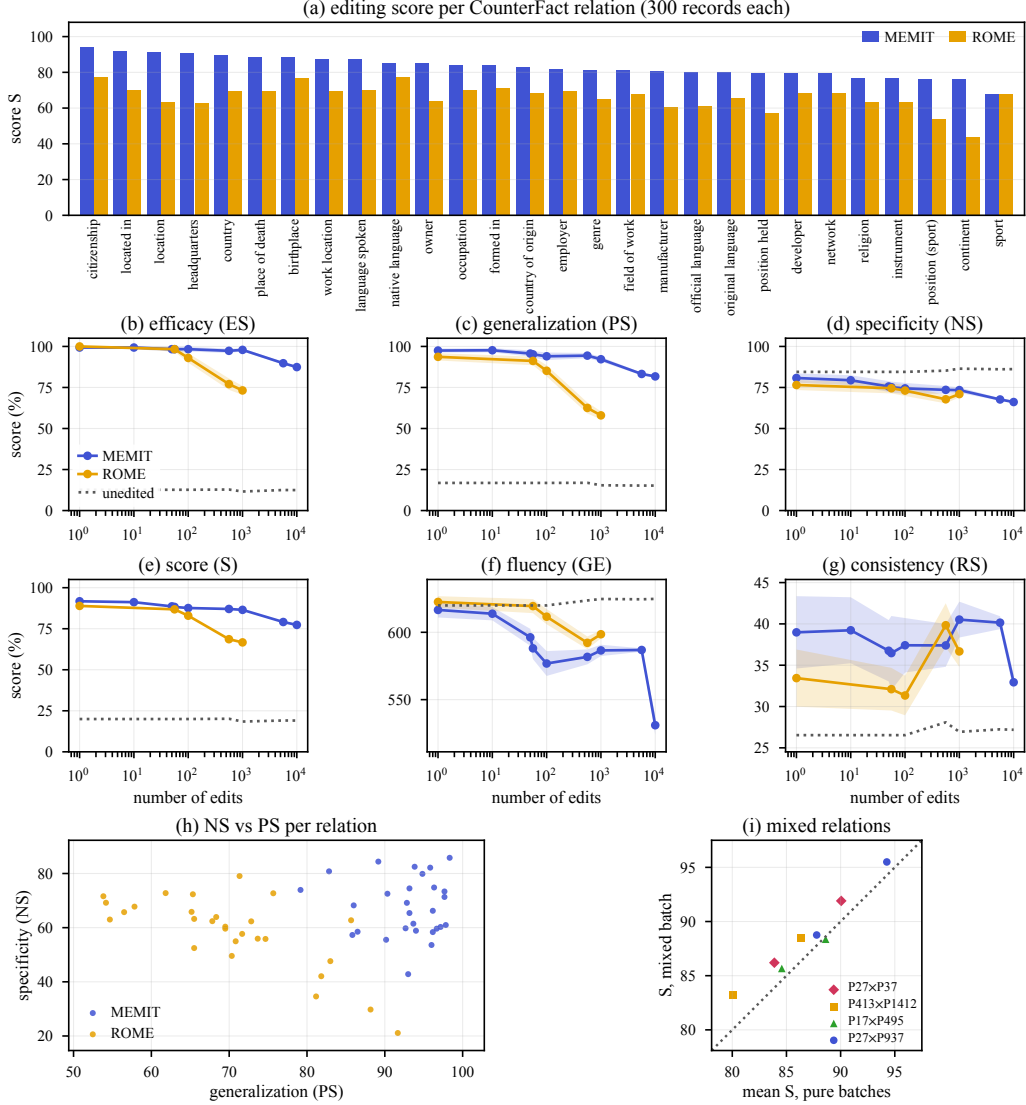


Figure 6: CounterFact editing results on Gemma-2-2B, for MEMIT applied to layers 3–10 and ROME applied to layer 6. Panel (a) reproduces Figure 6 of Meng et al. [2023b], panels (b)–(g) reproduce Figure 5, panel (h) is a second projection of the runs in panel (a), and panel (i) reproduces Figure 7. The metrics, from Meng et al. [2023a], are as follows: ES (efficacy) is the success rate of each edit; PS (generalization) is the success rate on paraphrase prompts; NS (specificity) is the success rate on unrelated facts; and S is the harmonic mean of ES, PS and NS. GE (fluency) is the weighted n -gram entropy of 100-token continuations from the generation prompts, which decreases as generated text becomes more repetitive. RS (consistency) is the TF-IDF cosine similarity between those continuations and reference text about the counterfactual object. (a) S for each of the 28 CounterFact relations with at least 300 records. MEMIT is applied as one batch of 300 edits per relation and ROME sequentially over the same 300. Six relations with fewer than 300 records are omitted (P138, P190, P264, P36, P407, P463). (b)–(g) Each metric against the number of edits applied to the model. MEMIT is evaluated at $n \in \{1, 10, 50, 56, 100, 562, 1000, 5623, 10000\}$ and ROME at $n \in \{1, 56, 100, 562, 1000\}$. Shaded regions are 95% confidence intervals over cases; dotted lines are the unedited model evaluated on the same records. (h) NS against PS for each relation, from the same runs as panel (a). (i) S of a batch mixing two relations against the mean S of the two single-relation batches, for the four relation pairs used in Appendix D of Meng et al. [2023b], at two batch sizes each. The dotted line is $y = x$. Two deviations from the upstream protocol apply: each point in (b)–(g) is evaluated on $\max(n, 300)$ records rather than 10,000, and the ROME curves stop at $n=1000$.

assigned by whichever criterion is satisfied first, with L the model depth:

$$\text{Role}(i) = \begin{cases} \text{Silent} & \text{no active probe (ours)} \\ \text{Dictionary} & \gamma_i \geq 0.80 \wedge \#\text{peaks}_i \leq 1 \\ \text{Say}(\cdot) & \Delta_{\text{fs},i} \geq 50 \wedge c_{\text{func}} \geq 0.90 \wedge \ell \geq \lceil \frac{7}{26} L \rceil \\ \text{Relationship} & \tilde{\text{sp}}_i < 0.45 \\ \text{Concept} & \ell \leq \lfloor \frac{3}{26} L \rfloor \vee c_{\text{sem}} \geq 0.50 \vee \Delta_{\text{fs},i} < 50 \\ \text{Review} & \text{otherwise (excluded)} \end{cases}$$

The layer cuts are their 7 and 3 indexed to Gemma-2-2B’s $L = 26$; on Gemma they resolve to exactly those integers. Functional tokens are identified by their fixed dictionary plus punctuation and a short-lowercase-alphabetic heuristic, and a functional peak is resolved to its semantic target by a directional search of at most seven tokens. Naming is role-specific: Dictionary/Concept features take their highest-activation semantic peak token; Say features are named Say (target); Relationship features are named (token) related over the admitted token set. Their probe-level stability filter (0.6) is described in their paper but absent from the released code that produced their tables; we default it off.

H Fine-Tuning the Cross-Layer Transcoder ($T_0 \rightarrow T_f$)

Experiment. We fine-tuned the CLT on M_e ’s activations over general Wikipedia text to obtain T_f . Results are in Figure 7: the replacement and completeness scores recover very little. This is largely a consequence of how localized the edit is: there is almost nothing global for the objective to correct. The feature agreement between A/B and A/C shows that the fine-tuned lens still fires on much the same features.

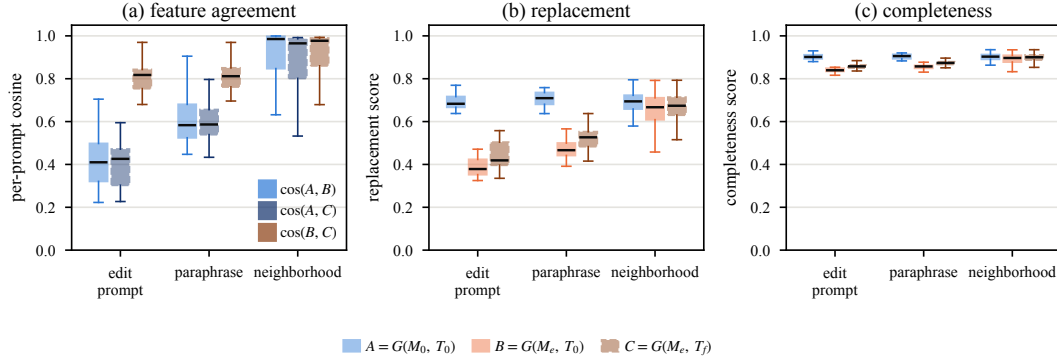


Figure 7: Fine-tuning the transcoder on the edited model does not undo the damage. Fine-tuning early-stopped at parity on all 20 trials after 181,454 tokens, so one edit does too little global damage for the objective to have anything to optimize against. The same three measures as Figure 1, with the adapted lens $C = G(M_e, T_f)$ added.

We describe our training regime below:

Parameterization. In order to train the adapted lens T_f , we never modify T_0 and instead train an additive delta over the frozen base, $T_f = T_0 + \Delta$, where Δ contains trainable corrections only on the slices a band- $\mathcal{B}$ edit most directly invalidates: decoder slices $\Delta W_{\text{dec}}^{(i \rightarrow l)}$ for every target layer $l \in \mathcal{B}$ and source layer $i \leq l$ (60 slices for $\mathcal{B} = [3, 10]$), plus encoder and bias deltas $\Delta W_{\text{enc}}^{(l)}, \Delta b_{\text{enc}}^{(l)}, \Delta b_{\text{dec}}^{(l)}$ for $l \in \mathcal{B}$, giving ≈ 2.6 B trainable parameters against the frozen 426k-feature base. All deltas initialize to zero, so training starts from $T_f \equiv T_0$ exactly; discarding Δ recovers the base CLT unchanged.

Objective and data. With M_e as a frozen data source (its forward pass runs without gradients; autograd touches only Δ), we minimize the band-restricted CLT loss $\mathcal{L}(\Delta) = \sum_{l \in \mathcal{B}} \|\widehat{\text{mlp}}_l -$

$\text{mlp}_l\|_2^2 + \beta \sum_{l \in \mathcal{B}} \|a^{(l)}\|_1$ where $\widehat{\text{mlp}}_l$ is T_f 's reconstruction of layer l 's MLP output on M_e , the L1 term is per-token normalized, and $\beta = 10^{-3}$ was frozen after a calibration benchmark chosen to keep the band's expected active features per token within $\sim 10\%$ of T_0 's. Training text is streamed from a 110k-row Wikipedia parquet (wiki_110k.parquet) under seed 7, with the rows of the wiki_v1 evaluation battery excluded so that training and evaluation stay separate; each row contributes BOS plus its first 128 content tokens, in batches of 16 rows with a seeded, per-epoch-reshuffled order.

Optimization. Deltas train in bfloat16 with 8-bit paged Adam at learning rate $\eta = 10^{-4}$; decoder sources above the band are parked on CPU between evaluations, which fits the run on a single 48 GB GPU.

Stopping rule. Training stops at *parity*: every 100 steps we evaluate CE_{rec} of T_f on M_e over held-out wiki rows, and stop once it reaches the value T_0 attains on M_0 (graph A 's faithfulness), with a hard ceiling of 30 M tokens. This rule is load-bearing for interpreting C : at $k = 1$, a single edit leaves global cross-entropy essentially intact, so *every* trial ($N = 20$) passed parity at the first evaluation: T_f is T_0 plus exactly 100 optimizer steps (181,454 tokens). This is a property of the stopping criterion, not of the optimizer failing to move: at larger k the parity target is not immediately reachable and training runs long (e.g. $\sim 16,600$ steps to the full 30 M-token ceiling). We report this because Section 5.1's comparisons between B and C inherit it: at $k = 1$, C is a minimally-perturbed T_0 , which is exactly the point: the edit gives the fine-tune almost nothing global to optimize against.

I CLT Faithfulness on Multi-Edit MEMIT

Figure 8 extends the single-edit comparison of §5.1 to k facts edited into one model, over the ladder $k \in \{1, 10, 50, 100, 562, 1000, 5623, 10^4\}$, and adds the control corner $D = G(M_0, T_f)$: the fine-tuned lens read back on the *unedited* model, i.e. what fine-tuning alone does absent any edit. Three readings. **(i)** The frozen lens degrades with k , and most of the damage is done by $k = 10$: on the edited prompt, B 's replacement score falls from 0.47 at $k = 1$ to 0.36 at $k = 10$ and then sits near 0.35 through $k = 10^3$, while completeness moves only from 0.85 to 0.82 and its active features per token rise $\sim 6.7\times$ over the same range (panel b): the error nodes absorb what the lens can no longer model. **(ii)** At $k = 1$, D coincides with A on every measure and $\cos(A, D) \approx 0.98$: fine-tuning alone changes nothing, consistent with the parity stopping rule of Appendix H. From $k = 50$ the fine-tune itself rewrites the lens (D 's CE_{rec} drops to ~ 0.4 and $\cos(A, D)$ to ~ 0.2 before both recover to ~ 0.75), so at large k the C -versus- B comparison is confounded by lens drift, not only by the edit; this is why the main text holds T_0 fixed and uses $k = 1$. **(iii)** The number of appeared nodes saturates near $k \approx 50$ while disappeared nodes grow slowly (panel h), and encoder rotation concentrates in the upper edited layers (panel i).

J Additional Ablation Results

Single-node patching (Figure 9), its dose-response over the λ grid (Figure 10), and the six-score comparison (Figure 11), referred to in §5.3. Table 3 defines the five scores compared there against $\phi^{(a)}$ of §4.1.

Table 3: The five further heuristic scores, ranked and patched alongside $\phi^{(a)}$ in Figure 11. Conventions as in §4.1. Note that $\phi^{(b)}$, $\phi^{(c*)}$ and $\phi^{(d)}$ each carry $|\delta_n|$ as a magnitude, as $\phi^{(a)}$ does, which is why they are not separable from it: $|\delta_n|$ is saturated at 1 for nearly every node on both sides.

Score	Name	Definition
$\phi_n^{(b)}$	Weighted influence	$ \delta_n \cdot \max(I_n^{(A)} , I_n^{(B)})^\alpha$
$\phi_n^{(c)}$	Influence z-score	$(\Delta I_n - \text{Med}(\Delta I)) / \text{MAD}(\Delta I)$
$\phi_n^{(c*)}$	Strict influence z-score	$ \delta_n \cdot \phi_n^{(c)}$
$\phi_n^{(d)}$	Influence-weighted edge movement	$ \delta_n \sum_{n'} \Delta w_{n \rightarrow n'} \max(I_{n'}^{(A)} , I_{n'}^{(B)})$
$\phi_n^{(e)}$	Edge redistribution	$\text{JSD}(\pi_n^{(A)} \ \pi_n^{(B)})$

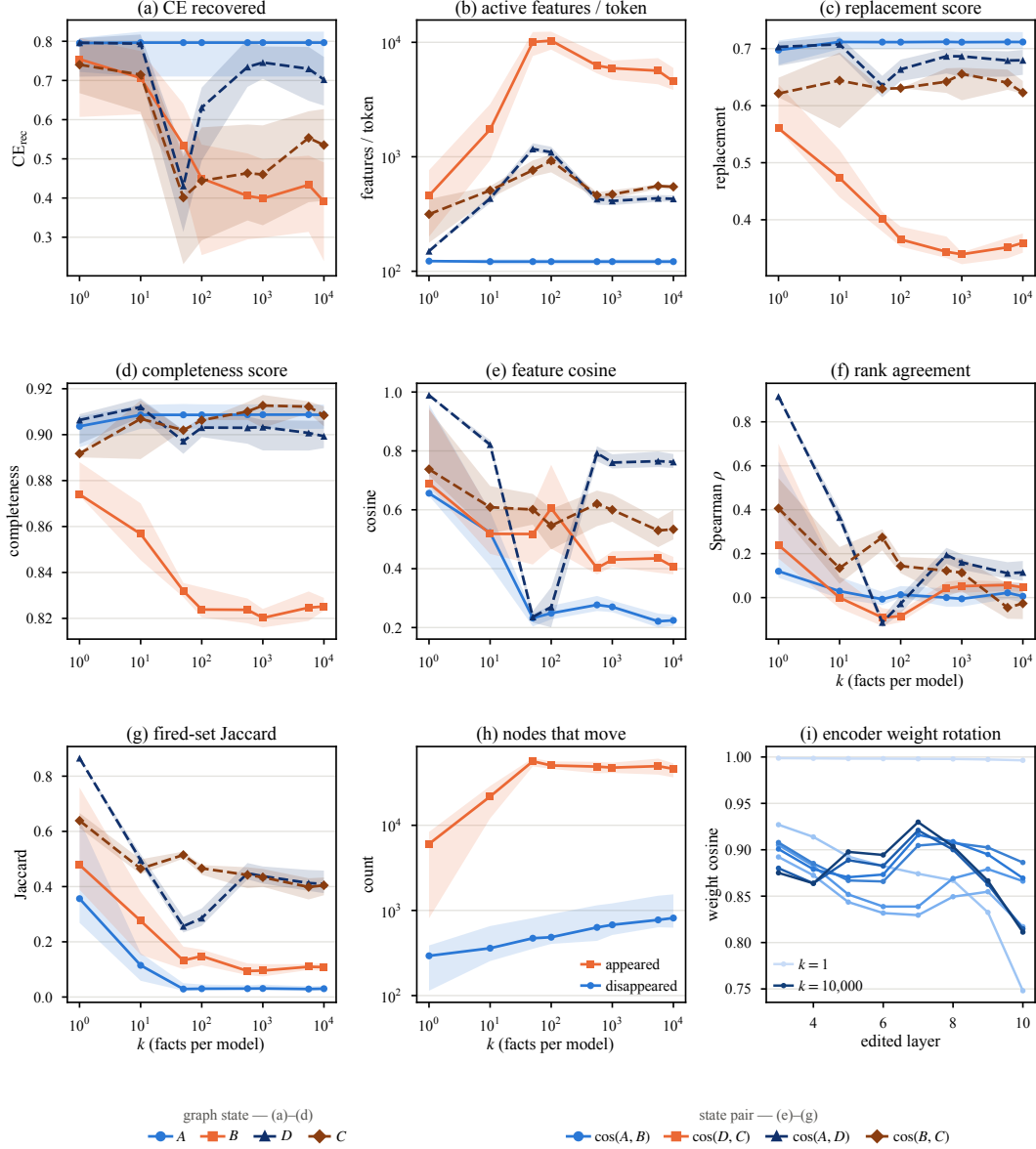


Figure 8: Multi-edit sweep across the full ladder ($k = 1$ to 10^4 facts per model). Panels (a)–(d) report the four graph states; (e)–(g) pairwise agreement, where solid lines isolate the edit’s effect (model changes, lens held) and dashed lines the lens’s effect (lens changes, model held); (h) the nodes that move; (i) encoder weight rotation per edited layer, one line per k . Lines are medians, ribbons the interquartile range. $k = 56$ is omitted: that run carries only the A/B arm.

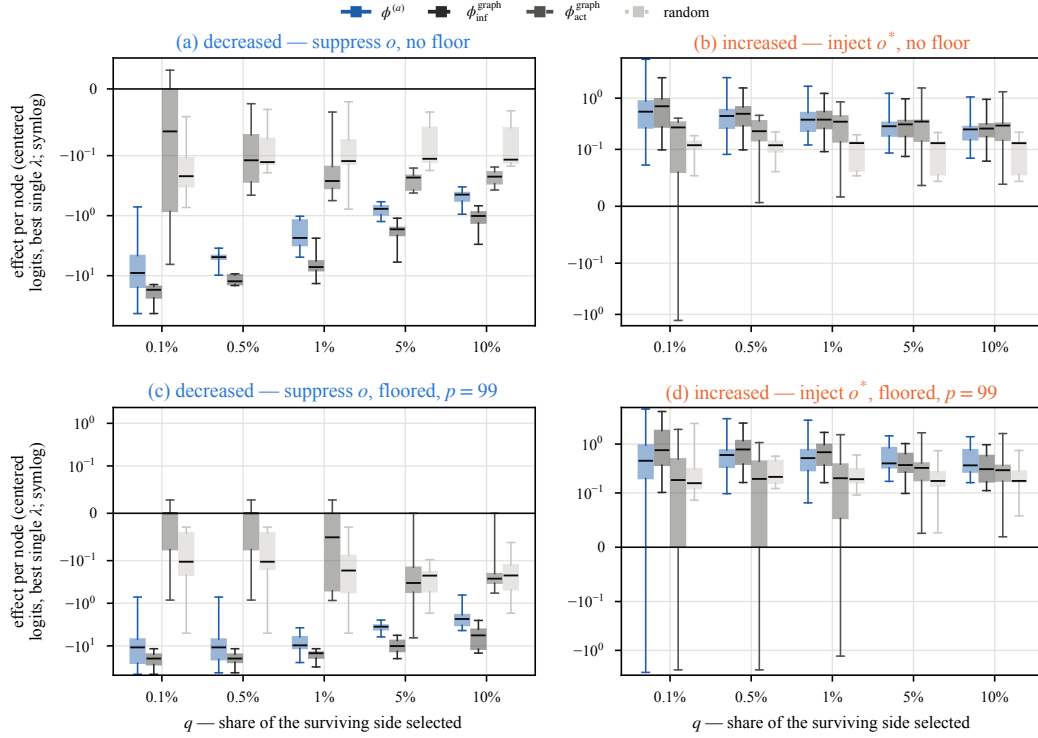


Figure 9: **Patching one node at a time, under both calibrations.** Protocol of §5.3; decreased nodes are scored against suppressing o , increased nodes against injecting o^* . Top row: no floor, q a share of the raw side. Bottom row: only nodes that passed the $p = 99$ null floor of Step 4, q a share of the survivors. Arms: $\phi^{(a)}$; the whole-graph baselines $\phi_{\text{inf}}^{\text{graph}}$ and $\phi_{\text{act}}^{\text{graph}}$, which rank *every* $t > 0$ node of that side’s graph by influence or activation on its own target; and a uniform random draw. Every arm selects $n = \lceil q |\text{side}| \rceil$ nodes; each node is read at its best single λ (dose–response in Figure 10), so levels are optimistic bounds. Boxes are the 8 CounterFact edits, whiskers full range, symlog axes; the full-depth window, where circuit-tracer also freezes LayerNorm, is excluded for all arms.

K Circuit-Diff Platform

K.1 Multi-Prompt Aggregation

To reduce the noise of relying on a single prompt, we hold the fact fixed and vary its phrasing over m prompts. Because token positions differ across phrasings, we track features rather than nodes. Within each prompt p we collapse a feature’s nodes into a single score $\Phi_i(p) = \sum_{n' \in T_i} \phi_{n'}^{(s)}$, where T_i is the set of nodes carrying feature i at any token position in p and s is the chosen score, normalize by that prompt’s median, and take the feature’s direction from the sign of its summed δ_i . We discard features whose movement is inconsistent across prompts, i.e. features with both positive and negative δ . We then pool across prompts with a power mean, $\text{MPD}_i = \left(\frac{1}{m} \sum_p \Phi_i(p)^\rho \right)^{1/\rho}$. We report $\rho = \frac{1}{2}$ as BREADTH (rewarding features that recur across phrasings) and $\rho = 2$ as PEAK (rewarding features that move strongly on any one phrasing).

K.2 Feature evidence for the case study: Eiffel Paris $\rightarrow$ Rome

The listing below is the evidence behind §6.1: the top-15 features per side under $\phi^{(a)}$ for the running example (each feature at its strongest node), with the tokens the feature promotes and its highest-activating dashboard context (peak token in brackets), read from the CLT’s released feature dashboards Piotrowski and Hanna [2025]. Nothing is labelled by hand or by a language model. Rows are in rank order within each side; §6.1 (*Beyond the token*) reads the Roman-history, mythology and

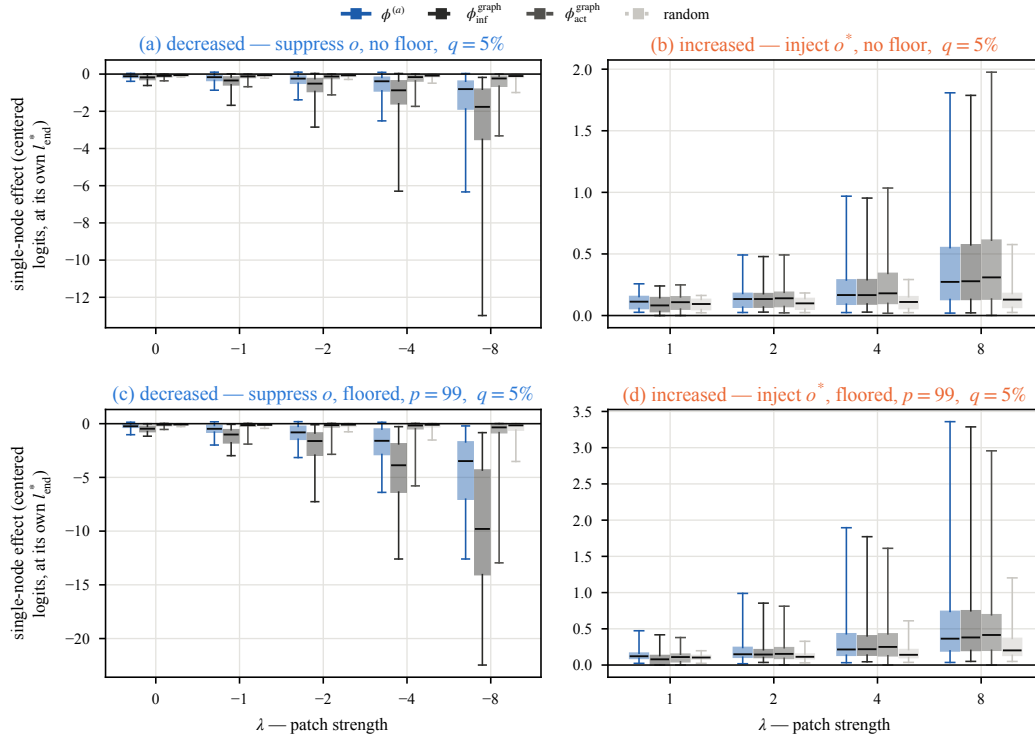


Figure 10: **Dose-response of single-node patches at $q = 5\%$.** Same protocol, arms and edits as Figure 9; top row unfloored, bottom row restricted to the $p = 99$ survivors. The grids differ by side: suppression sweeps $\lambda \in \{0, -1, -2, -4, -8\}$ (0 is the pure ablation), injection $\lambda \in \{1, 2, 4, 8\}$. Boxes pool nodes over 8 CounterFact edits, whiskers 5th–95th percentile.

Church features on the appeared side and the French-history features on the disappeared side off this listing.

Dagger n marks a feature that is in the per-feature top-15 of either side in $n \geq 3$ of the 25 pooled edits (the running example plus the 24 CounterFact edits of the sweep in §5.3. The count is therefore over the whole pool, so a feature may carry a dagger without every one of its other appearances being printed here; logit tokens in non-Latin scripts are skipped (count in the generated file’s header). Generated by `cs print-cards` from the same `scores.py` ranking the viewer uses; the same command prints the per-edit listing for any subset of the pool.

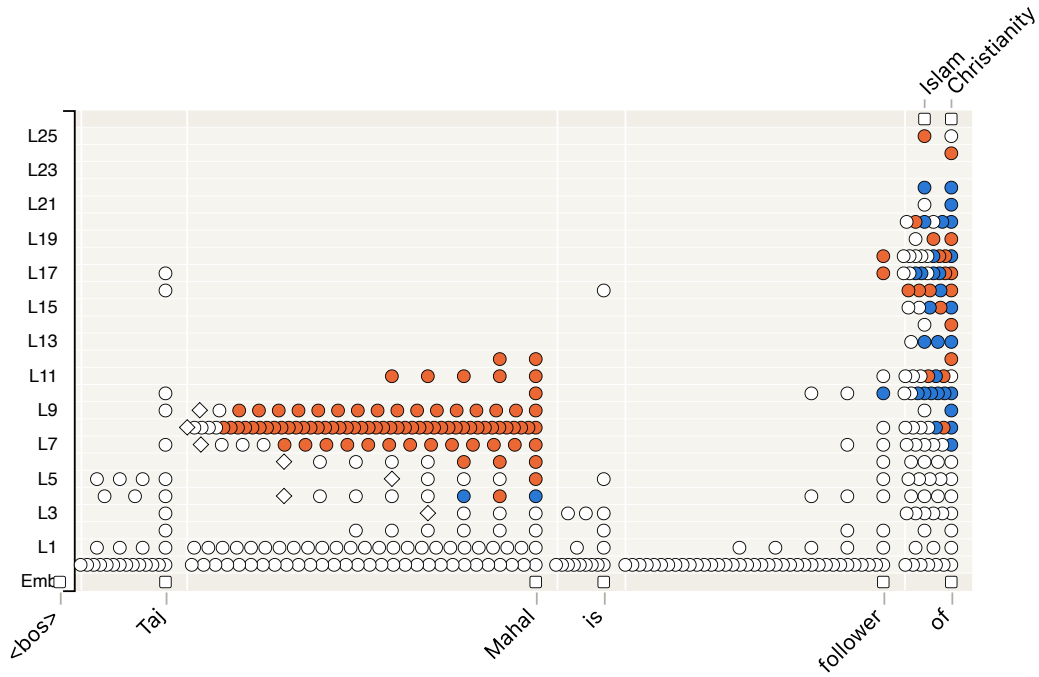
Side (based on <i>A</i>)	Feature	Top logits	Context
<i>Eiffel Paris</i> → <i>Rome</i>			
disappeared	L18/f7269 ^{†4}	le, Ê, Ré	Museum at London, Grand Palais[at] Paris, National
disappeared	L20/f9908	normaux, TagMode, financières	be seen at The Louvre Museum[in] Paris, France
disappeared	L18/f14136 ^{†3}	WithFormat, addCriterion, uska	of Brassempouy[in] 18
disappeared	L25/f9331	tvguidetime, stratégique, nôtre	Supérieure, Paris, France[,] 3 Department
disappeared	L19/f472 ^{†3}	departments, department, Giver	by the events of 1[7]89,
disappeared	L20/f9226	Valera, Butte, gehör	Lachaise Cemeteray[in] Paris.¶
disappeared	L12/f6700	llamado, stället, antaranya	on National Park on the island[of] Java, Indonesia
disappeared	L15/f8344	complexContent, Manhattan, ///<>	kickoff event at the Franklin Institute[in] Philadelphia today
disappeared	L16/f12025	ardless, enumi, RectangleBorder	showcased at the ceremony, Mayor of[Jerusalem] Nir Barkat
disappeared	L21/f6014	Alsace, departament, iredo	had advanced as far as Alsace[-]Lorraine, provinces
disappeared	L7/f803	Italijani, ItemBackground, ++	the middle one of which,[built] of stone,
disappeared	L20/f9342	#####., expandindo, estekak	He was superior of the seminary[of] Issy when

Side (based on A)	Feature	Top logits	Context
disappeared	L16/f14688	Manbalar, Geplaatst, AssemblyTitle	Gemeinden in Germany...[in[the] Ain[Ain
disappeared	L19/f12469 ^{†3}	Havre, Lyons, Constant	ote at Marseilles—He[reaches] Bombay—Letter
disappeared	L15/f4283	<eos>, ,	her arrangement with Hades regarding her[daughter] Persephone.
appeared	L20/f4888	HomeAsUp, Lati, +##	from Ascoli Satriano[in] Puglia with a
appeared	L7/f12622 ^{†12}	pu, rap, inge	that divides the waters of French[Broad] and Big Pigeon
appeared	L19/f4224	Caesar, Fla, legion	and on the mother's[from] Trajan,
appeared	L17/f13216	Italij, Italians, Italy	India, France, Greece,[Italy], and the
appeared	L20/f7760	Eternal, Umb, eternal	the village of Acquapendente[in] Tuscany by Julian
appeared	L20/f781	XCTest, multer, coscienza	National Laboratory of the Italian Institute[of] Nuclear Physics,
appeared	L18/f802	Accesat, Ple, yntaxException	So for example, all Romans[loved] to see people
appeared	L8/f14670	##, Roskov, GEBURTSDATUM	· Washington, DC 2[0]201
appeared	L20/f13965	Split, Piran, Cef	usual big cities and spend Halloween[in] Transylvania! Con
appeared	L20/f626	R, first, Rk	the proclaimed 'wealthiest man[in] Rome', was
appeared	L18/f6681	Greco, temples, pyramids	. But it is also home[to] vast archeological ruins
appeared	L8/f14633	ItemBackground, GraphicsUnit, VideoCapture	Roberts, a biological anthropologist with[Glasgow] University's
appeared	L22/f4296	av1, R, szta	17)—rival popes[in] Rome and Avignon
appeared	L16/f16014	Bologna, Padua, Ferrara	other Hebrew presses were set up[in] Mantua,
appeared	L19/f8642	skydd, Maggio, Repubblica	. [Leonardo Campanelli of] the] University of Ferrara

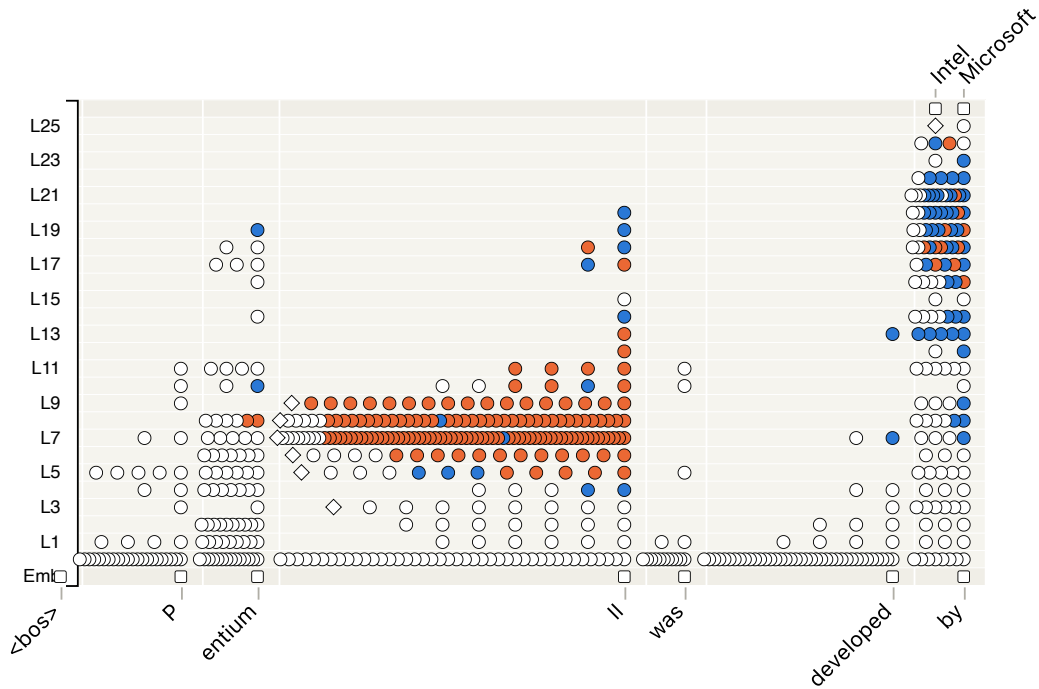
K.3 Two further edits: Taj Mahal and Pentium II

The evidence behind §6.2, in the same form and from the same generator as Appendix K.2: the circuit-diff of each edit (Figure 12) and the top-6 features per side under $\phi^{(a)}$ (listing below). Both edits are from the 24-edit sweep of §5.3 (CounterFact cases 21243 and 6467); daggers are counted over the same 25-edit pool.

Side (based on A)	Feature	Top logits	Context
<i>Taj Mahal Islam → Christianity</i>			
disappeared	L22/f3758	Shah, Allah, Arjuna	children. It is a story[of] a young boy
disappeared	L25/f9787 ^{†11}	auroit, varandra, feroit	flame. In the midst of[this] turbulence,
disappeared	L20/f65	Mulai, Cali, Mih	of Spanish life has a touch[of] Islam. Spanish
disappeared	L13/f5332	Autoritní, BoxFit, myself	<bos>The Leviathan[of] Parsonstown:
disappeared	L7/f3997	must, Efq, chrétien	<bos>Jesus —[Our] Shepherd[Week
disappeared	L20/f7377	Portale, LookAnd, ostavi	Kalam was closely allied with[Islamic] jurisprudence and typically
appeared	L17/f11321	Protestant, Christian, churches	them - tend not to be[regular] church-goers
appeared	L18/f842	Roman, Lutheran, Roman	-denial traditionally observed by Catholics[and] some Protestant denominations
appeared	L21/f10906	practising, faith, Belie	prior to Christianity becoming an officially[tolerated] religion of the
appeared	L25/f12909 ^{†4}	. }~\, ividual, hematical	yscale and contrast ratio mean basically[the] same thing.
appeared	L6/f11532	PerformLayout, \$\\, \$\\\$	<bos>Summary[:] As part of
appeared	L15/f5748	Judaism, Byzantium, Arabs	fine arts developed in strong interaction[with] European art,
<i>Pentium II Intel → Microsoft</i>			
disappeared	L20/f13952	iVar, BorderRadius, Centr	processors or a Hyper Threading[enabled] CPU can be
disappeared	L20/f8977	poveznice, tagHelperRunner, ConstraintMaker	relative quietness, the Taiwanese[manufacturer] Winbond recently
disappeared	L17/f13795	architectures, architecture, archite	, Intel's x8[6] processor architecture is
disappeared	L18/f10518	stateProvider, IConfiguration, MOD0	CL, and updated versions of[Intel]'s Fortran
disappeared	L18/f909	Think, kt.r, offsetof	to _OS is that it[']s a single
disappeared	L14/f1292	SAP, Intel, Microsoft	as to persuade rivals Novartis[and] Sanofi to
appeared	L18/f4221	pherson, Chwiliwch, StructEnd	use in the report, as[well] as calculated fields
appeared	L5/f11228	iStock, bootstrapcdn, These	KEY constraint using Transact-[SQL] or SQL-
appeared	L13/f6564	AssemblyCulture, AddTagHelper, EndProject	Journal of Environmental Research and Public[Health]. In this
appeared	L10/f12206	Windows, Microsoft, Microsoft	on missions beyond Earth orbit.[]Well, this
appeared	L8/f14613	purpose, itself, Jefus	by creating a simple VB Windows[form] project, and
appeared	L7/f3241	jwt, cherc'hés, propOrder	from both human rights groups and[technolog]ists.[]



(a) *Taj Mahal is follower of: Islam → Christianity*



(b) *Pentium II was developed by: Intel → Microsoft*

Figure 12: **The diff on two further edits.** Conventions as in Figure 4(a, b): columns are token positions, rows are layers, open markers are nodes present in both graphs, and filled markers carry the diff's verdict: **blue** = disappeared, **orange** = appeared. The labelled columns at the right edge are the two answer logits. Both edits show the signature of §6.1: appeared nodes form a band across the MEMIT layers at the subject's last token (*Mahal*, *II*), and the final position carries the readout of the old and new answers.